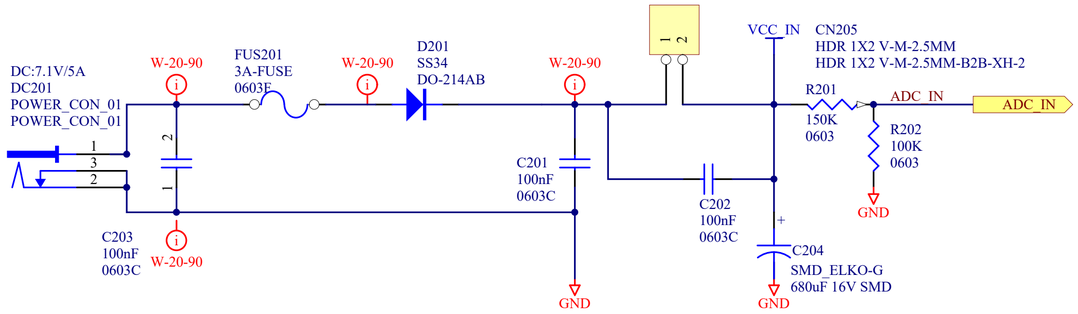
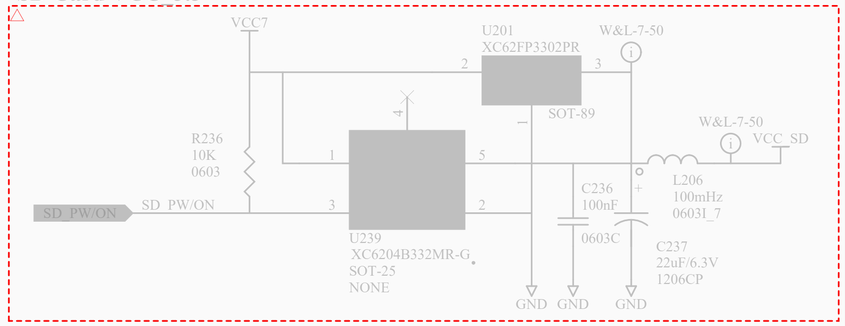
POWER

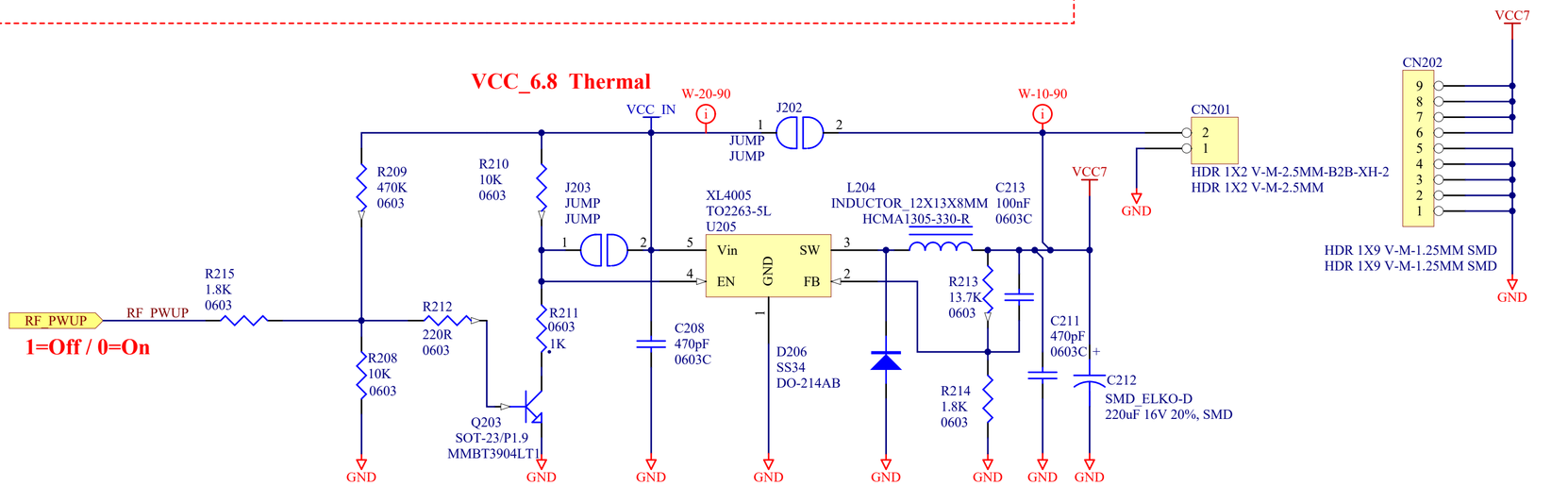
VCC IN 6.8V to 12V



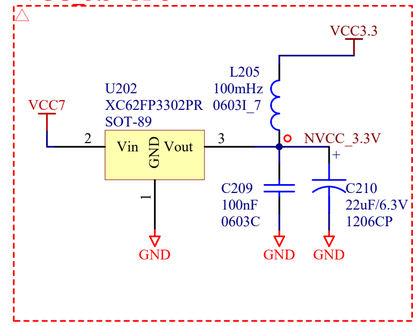
SD Card VCC_3.3



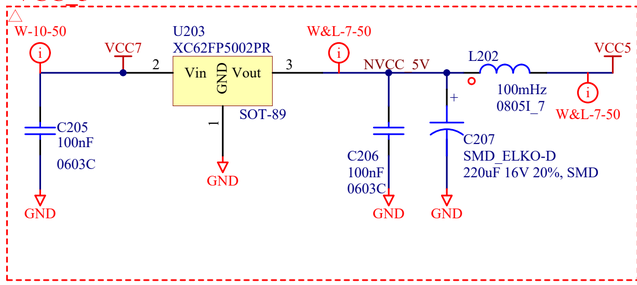
VCC_6.8 Thermal



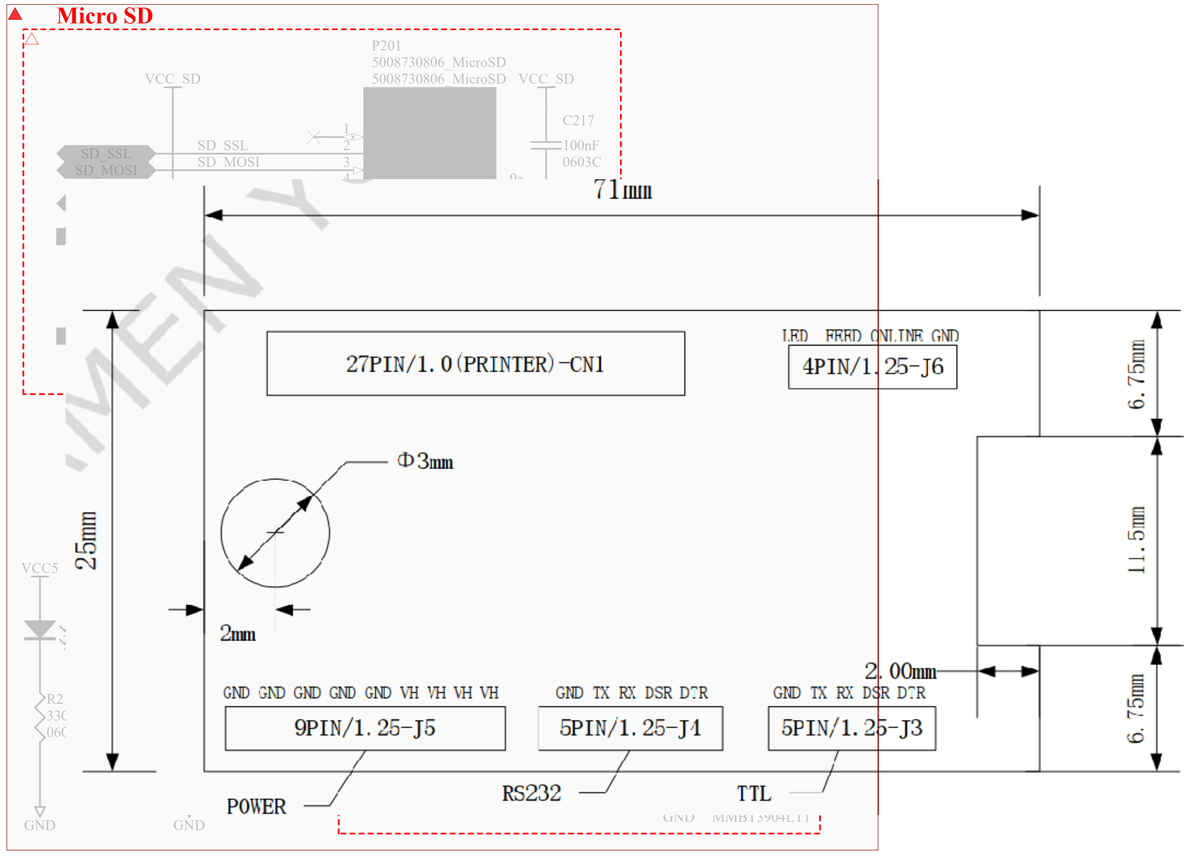
VCC_3.3 CPU



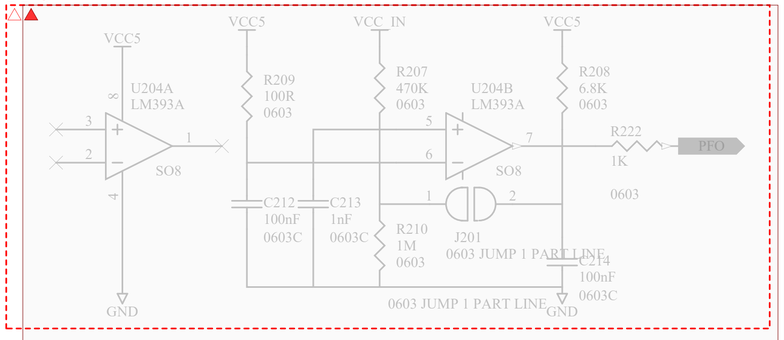
VCC_5



Micro SD



PFO



Title POWER - KEYBOARD			DENSI
Size:	Number:2	Revision: 08-07-2009	ESP8266_Wifi_POWER_260720.SchDoc ESP8266_Wifi_260720.PcbDoc ESP8266_Wifi_POWER_260720
Date: 7/27/2026	Time: 7:49:52 PM	Sheet 1 of 4	
File: C:\Users\yguna\Documents\Proje\Remote Control\ESP8266_Wifi_Pos_260720\ESP8266_Wifi_POWER_260720.SchDoc			

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```
/*  ESP8266
  1- Reset
  2- ADC                    //Button Analog 4 Key
  3- Enabled
  4- IO16  BT SET           //  Button3  SETUP  <- TM1638(LOAD)
  5- IO14  BT M+ PRINT      //  Button2  PRINTM+
  6- IO12  Buzzer           //  Buzzer
  7- IO13  Paper Control    //  Paper Control Interupt pini
  8- VCC
  15- GND
  16- IO15 -                //  TM1638(SDI)
  17- IO2  - TX1            = Uart2 To RP80
  18- IO0  - BT PRINT       //  Button 1 PRINT  <- TM1638(SCLK)
  19- IO4  - RTC(SDA)       = LCD = RTC
  20- IO5  - RTC(SCL)       = LCD = RTC
  21- IO3  - RX0            = Indicator
  22- IO1  - TX0            = Indicator
*/

#define TRIGGER_BtPrint      0      //12
#define TRIGGER_BtMPlus      14     //2
```

4 KEYBOARD & Display PIN INPUT

VCC7, VCC5, JP106, CN103 FPC-12 V FPC-12 V

IO16_SET, IO0_PRINT, IO14_M+, PRN_FEED, SDA, SCL

SET, PRINT, M+, FEED, BUZZER

0603 JUMP 2 PART LINE

OLED Display out SCL_SDA

VCC3.3, CN105, SCL, SDA, RED, GREEN, WHITE, BLACK, SIP4-F-2MM, SIP4-F-2MM, GND

RS232

JP101 0603 JUMP 2 PART LINE, 0603 JUMP 2 PART LINE, VCC3.3 VCC5

HOL-60, HOL106 HOL-60 NONE, HOL107 HOL-60 NONE, GND

L102 100mHz NONE 0603I_7

CN102 RED GREEN WHITE BLACK, SIP4-F-2MM SIP4-F-2MM

CN102 RED GREEN ------- WHITE BLACK, JST_B05B-SZ

TXD0, RXD0, TXD1, GND

EM205 BOARD KEY INPUT

GND, CN106, PRN_ONLINE, PRN_FEED, PRN_LED, SIP4-F-2MM, SIP4-F-2MM

OPTIONAL 3KEY

CN110, IO16_SET, IO0_PRINT, IO14_M+, PRN_FEED

TER-BLOCK 5P, TER-BLOCK 5P, TER-BLOCK 5P, GND

CN113

VCC5, CN113, FPC-8 V, FPC-8 V

FT800_CS, FT800_MISO, FT800_MOSI, FT800_SCK, FT800_INT

DP3_15, BUZZER, PRN_LED, IO14_M+, IO16_SET, FT800_PD

CN403

VCC5, R403 1.8K 0603, GND

FT800_INT, FT800_SCK, FT800_MOSI, FT800_MISO, FT800_CS

CN403, TO_Main

5- IO16_SET, 4- IO0_PRINT, 3- IO14_M+, 2- PRN_FEED

REAL TIME CLOCK

DC:7.1V/5A

VBAT, R102 470K 0603, R101 100R 0603, D101 LL4148 1206D, VCC5

CU101 100nF 0603C, R110 1.5M 0603, JM101 JUMP JUMP

BATTERY_BS-1220, BATTERY BS-1220-2AU

B102, HDR 1X2 V-M-2MM, SIP2-F-2MM, SIP2-F-2MM

B101, NONE, 2 VCC

U102

VCC3.3 VBAT

32768HZ, XTAL_120X400, Y101

SCL, SDA, OSC32_IN, OSC32_OUT

SCL, SDA, X1, X2, VCC, VBAT, SQW/OUT, GND

C102 22pF 0603C, C103 22pF 0603C, GND, GND

DS1307Z, SO8

OPTIONAL ADC

VCC3.3, L101 100mHz NONE 0603I_7

CN101, ADC_IN, SIP3-F-2MM, SIP3-F-2MM, GND

BUZZER

VCC5, L110 100mHz 0603I_7, R112 10K 0603, R111 10K 0603

BUZZER, BZ101 BUZZER 12X10mm BUZZER_0.5

R113 4.7K 0603, Q101 SOT-23/P1.9 MMBT3904LT1, GND

VCC3.3CPU

VCC3.3CPU, R103 27K 0603, D102 LL4148 1206D, RST

SW101 NONE, CU102 100nF 0603C, GND, GND

SW-TACK SKRG, SW-TACK SKRG

L103

VCC3.3, L103 100mHz 0603I_7, VCC3.3CPU

C110 22uF/6.3V 1206CP, GND

R108 10K 0603, R105 10K 0603

M101, AI

RST, ADC_IN, IO16_SET, IO14_M+, BUZZER, PRN_LED

RST, ADC, EN, IO16, IO14, IO12, IO13, VCC, CS0, MISO, IO9

IO1/TX0, IO3/RX0, IO5, IO4, IO0, IO2/TX1, IO15/RTS, GND, SCLK, MOSI, IO10

TXD0, RXD0, SCL, SDA, IO0_PRINT, TXD1, DP3_15

ESP8266_ESP12F, ESP-12

ADC_IN, ST_CP_S, RF-SCLK, RF-MISO, RF_MOSI, FT800_CS

SH_CP_S, RF_CSN, RF_CSN

SW103, SW-TACK SKRG, NONE, SW-TACK SKRG

GND, GND, GND

VCC3.3

VCC3.3, NONE, R107 10K 0603

JM102, JUMP JUMP, RF_PWUP, RF_PWUP

R106 1.8K 0603, GND

L104

VCC5, L104 100mHz 0603I_7 NONE

100mHz, 100mHz, 0603I_7, PC_TX1, PC_RX1

L105, NONE, L106 NONE

P101, PC-TX, PC-RX, BU_TEL6HP, BU_TEL6HP, USED

GND

L107

VCC5, L107 100mHz 0603I_7 NONE

100mHz, 100mHz, 0603I_7, PC_TX0, PC_RX0

L108, NONE, L109 NONE

P102, PC-TX, PC-RX, BU_TEL6HP, BU_TEL6HP, USED

GND

JM103 / JM104

PRN_FEED, PRN_LED

JM103, JM104, JUMP JUMP, JUMP JUMP

TXD1, JP105, 0603 JUMP 2 PART LINE

EM_DTR, EM_DSR, EM_RX, EM_TX, TXD1_#

CN104, BLUE, GREEN, YELLOW, RED, BLACK

HDR 1X5 V-M-1.25MM SMD, HDR 1X5 V-M-1.25MM SMD

GND

LCD

LCD CABLE PCB INSTALLATION

VCC5, CN608, VCC5_L

RandomNerdTutorials

RST, ADC0, ENABLE, WAKE, SCLK, MISO, MOSI, VCC

GPIO16, GPIO14, GPIO12, GPIO13

GPIO1, GPIO3, GPIO5, GPIO4, GPIO0, GPIO2, GPIO15, GND

TXD0, RXD0, SCL, SDA, FLASH, TXD1, CS, TXD2

GPIO11, GPIO7, GPIO9, GPIO10, GPIO8, GPIO6

SDCMD, SDD0, SDD2, SDD3, SDD1, SDCLK

RS232

C104 1uF 0603C, C105 1uF 0603C

U101SO16, C1+, C1-, C2+, C2-, VDD, VCC, T1IN, T2IN, T1OUT, T2OUT, R1OUT, R2OUT, R1IN, R2IN, GND, VEE

VCC5, C106 1uF 0603C, C107 1uF 0603C, GND

TXD0, TXD1_#, RXD0, RXD1, PC_RX0, PC_RX1, PC_TX0, PC_TX1

C108 1uF 0603C, CU105 100nF 0603C, VCC5, GND, GND, GND

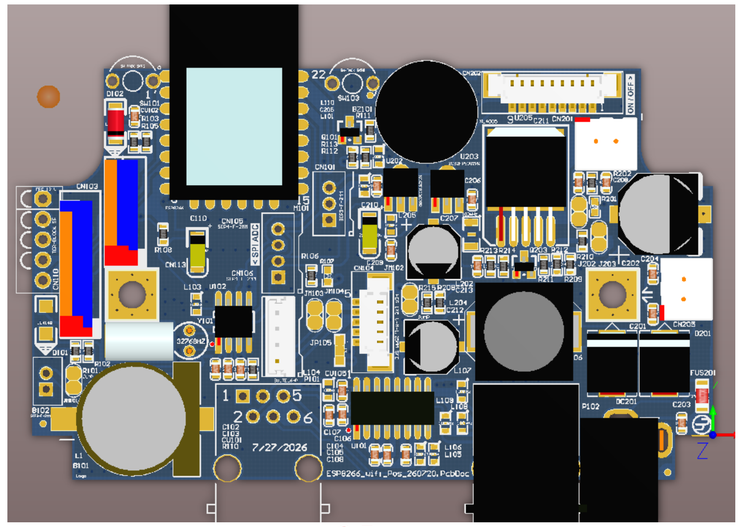
MAX232ACSE

1. SERIAL = BOOT & SCALE

Title MAIN BOARD CPU			DENSI
Size:	Number:3	Revision: 08-07-2009	ESP8266_Wifi_CPU_260720.SchDoc ESP8266_Wifi_260720.PcbDoc ESP8266_Wifi_CPU_260720
Date: 7/27/2026	Time: 7:49:52 PM	Sheet 2 of 4	Cannot open file C:\Users\Yuna\Documents\Proje\
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ESP8266_Wifi_Main_260720.SchDoc



ESP8266_Wifi_POWER_260720
ESP8266_Wifi_POWER_260720.SchDoc

ESP8266_Wifi_CPU_260720
ESP8266_Wifi_CPU_260720.SchDoc

ADC_IN
RF_PWUP

ADC_IN
RF_PWUP

ESP8266_BUTTON4_260720
ESP8266_BUTTON4_260720.SchDoc

MOUNTING_HOLE_3.5MM
NONE
HOL15
MOUNTING_HOLE_3.5MM

MOUNTING_HOLE_3.5MM
NONE
1
HOL16
MOUNTING_HOLE_3.5MM

MOUNTING_HOLE_3.5MM
NONE
HOL17
MOUNTING_HOLE_3.5MM

MOUNTING_HOLE_3.5MM
NONE
1
HOL18
MOUNTING_HOLE_3.5MM

TOP LAYER

HOL5 HOL6
HOL-TOP-30 HOL-TOP-30
HOL9 HOL-TOP-30
HOL-TOP-30 HOL-TOP-30
HOL-TOP-30
NONE
NONE
NONE

BOTTOM LAYER

HOL7 HOL8
HOL-BOT-30 HOL-BOT-30
HOL10 HOL-BOT-30
HOL-BOT-30 HOL-BOT-30
HOL-BOT-30
NONE
NONE
NONE

L1
DENSI
Logo
NONE
LOGO Densi

CURSOR_09
ESP8266_Wifi_Main_260720.SchDoc

PCB_Prn_LPC176X_210705
USED

```
ESP8266_Wifi_260720

ESP8266_Wifi_Main_260720.SchDoc                1       101     4 - Block Diagram
ESP8266_Wifi_POWER_260720.SchDoc               3       301     4 - Power
ESP8266_Wifi_CPU_260720.SchDoc                 2       201     4 - CPU
ESP8266_Wifi_BUTTON4_260720.SchDoc               4     401     4- Printer -SPI

*********************************************************************
ESP8266_BUTTON4_260720.PcbDoc                  Keyboard Pcb

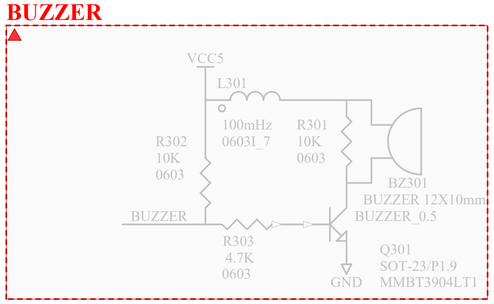
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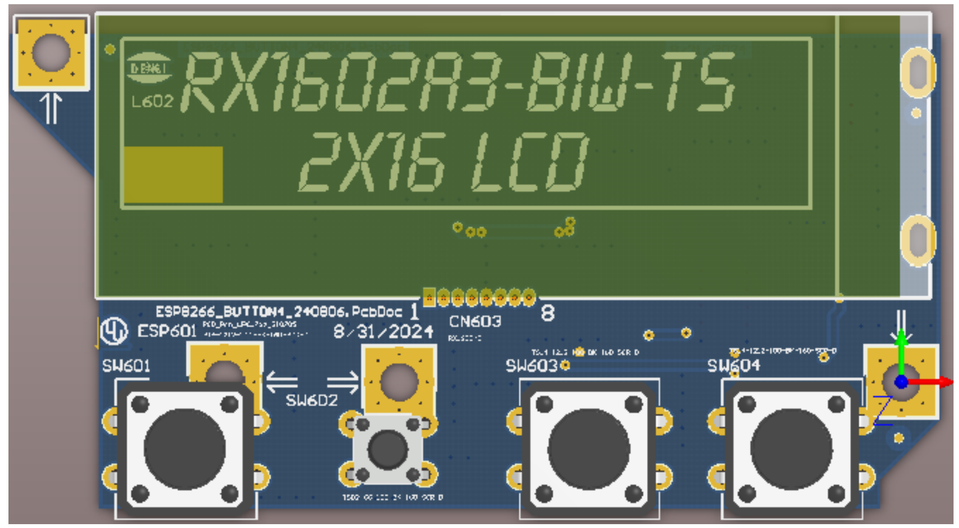
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Size:	Number:1	Revision: 08-07-2009	ESP8266_Wifi_260720.PcbDoc ESP8266_Wifi_Main_260720
Date: 7/27/2026	Time: 7:49:52 PM	Sheet 3 of 4	
File: C:\Users\yguna\Documents\Proje\Remote Control\ESP8266_Wifi_Pos_260720\ESP8266_Wifi_Main_260720.SchDoc			

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BUZZER

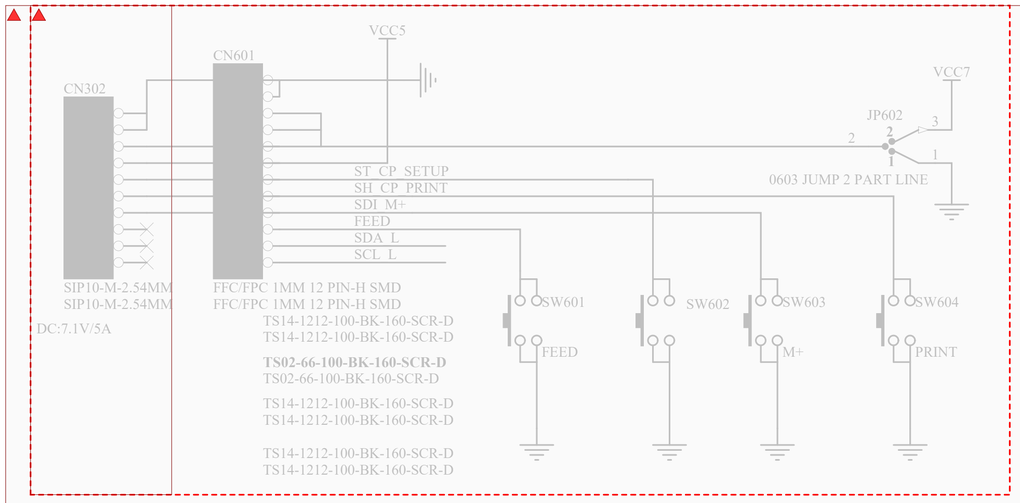




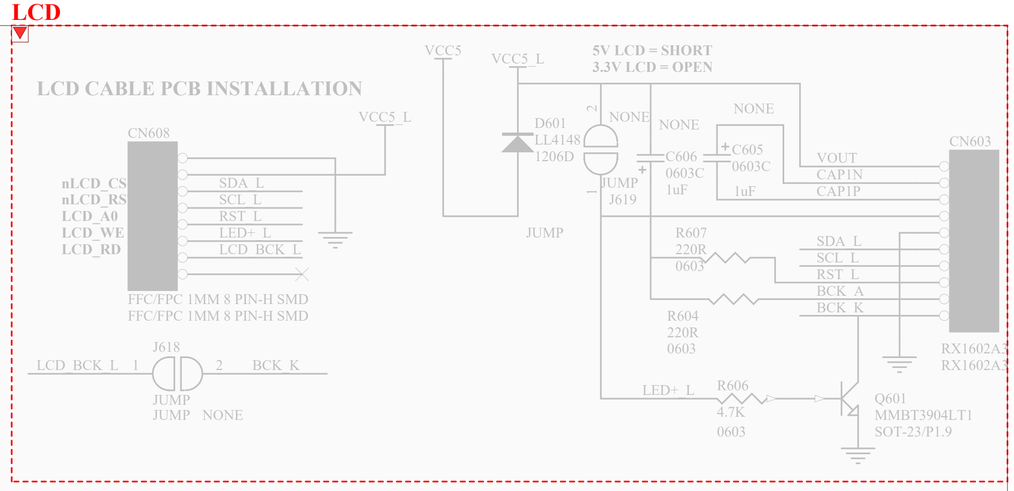
43.00mm

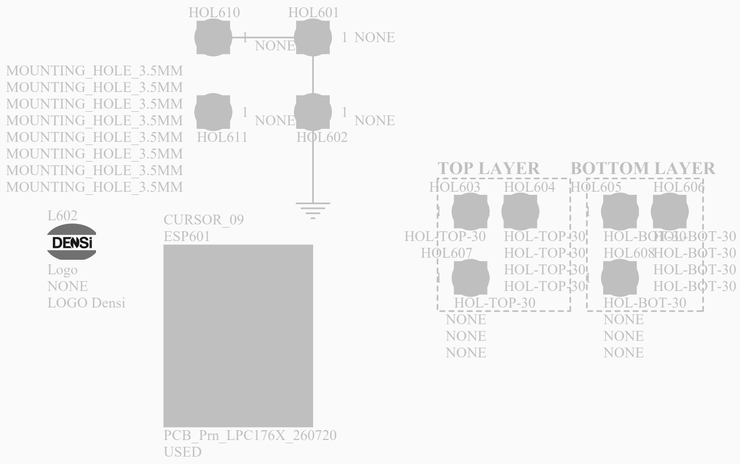
83mm

12x12x9mm BUTTON



LCD

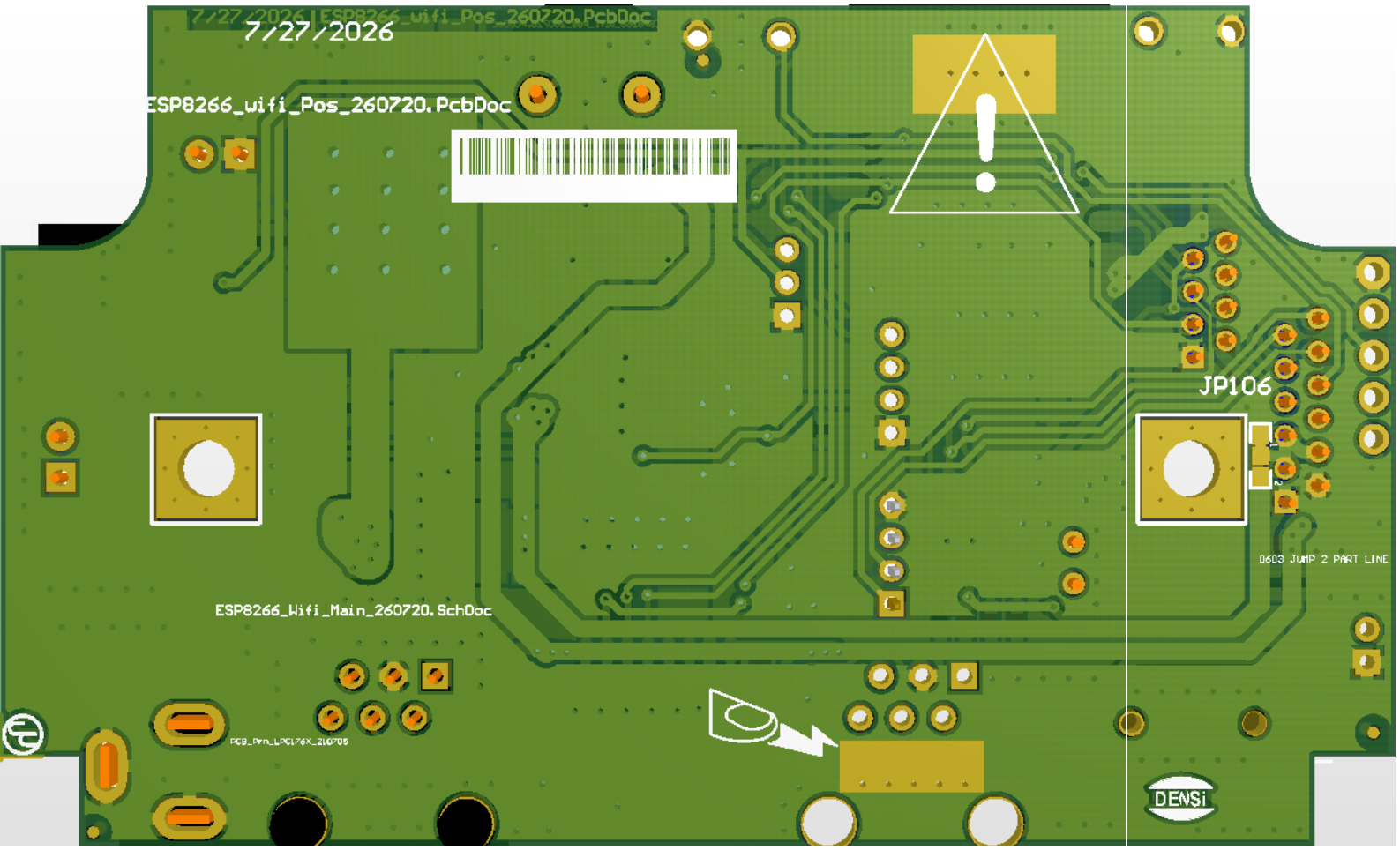


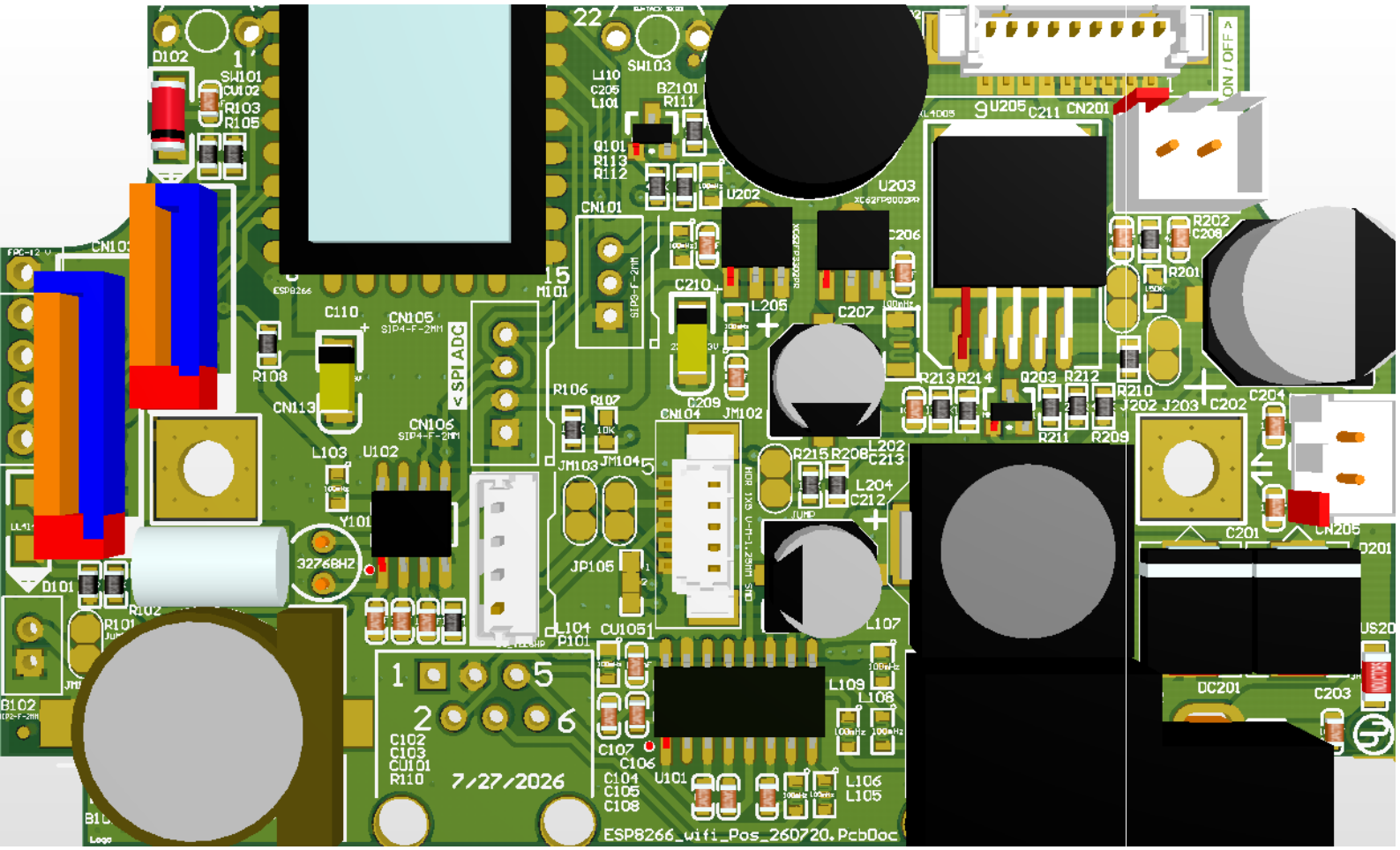


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Size:	Number: 4	Revision: 08-07-2009	ESP8266_Wifi_POWER_260720.SchDoc ESP8266_Wifi_260720.PcbDoc ESP8266_Wifi_POWER_260720
Date: 7/27/2026	Time: 7:49:52 PM	Sheet 4 of 4	
File: C:\Users\yguna\Documents\Proje\Remote Control\ESP8266_Wifi_Pos_260720\ESP8266_BUTTON4_260720.SchDoc			

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7/27/2026 ESP8266_wifi_Pos_260720.PcbDoc
7/27/2026
ESP8266_wifi_Pos_260720.PcbDoc
JP106
0603 JUMP 2 PART LINE
ESP8266_Wifi_Main_260720.SchDoc
DENSi


D102
SW101
R103
R105
SW103
L110
C205
L101
BZ101
R111
Q101
R113
R112
U202
U203
U205
C211
CN201
ON / OFF
CN101
C206
R202
C208
R201
C210
L205
C207
R213
R214
Q203
R212
R210
J202
J203
C202
C204
C110
CN105
SPI ADC
R108
CN113
CN106
R106
R107
10K
JM102
JM103
JM104
L103
U102
Y101
32768HZ
R215
R208
L202
C213
L204
C212
C201
CN205
D201
D101
R102
R101
JP105
L104
P101
CU105
L107
L109
L108
US20
DC201
C203
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CU101
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7/27/2026
C107
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L105
ESP8266_wifi_Pos_260720.PcbDoc